CHANDERNAGORE COLLEGE

SEMESTER – II B.Sc. (MINOR) INTERNAL ASSESSMENT EXAMINATION, 2024

SUBJECT : PHYSICS COURSE CODE : PHYS2021 COURSE NAME : MECHANICS

FULL MARKS: 10 TIME: 30 MINUTES

ATTEMPT ANY 02 OUT OF THE FOLLOWING 03 QUESTIONS

2 x 5 = 10

Q1 : Define inertial frame of reference. Consider a rocket system of mass M moving with velocity **V** with respect to an inertial frame O at time t. Between time t and t + δt, a mass δm is expelled with a velocity **u** w.r.t. O. During this interval, velocity of the system of mass M-δm changes to **V**+δ**V** w.r.t. O. Find the rate of change of momentum in the limit δt→0 in terms of M, **V** and **u**.

Q2 : State the principle of conservation of linear momentum. Define the impulse of a force. locate the centre of mass of a "T" shaped structure as shown in the figure: [2+3]

Q3 : Derive the expression of the gravitational potential due to a point mass m at a distance of r away from it.